

* Syntax:-

→ syntax is the set of rules used to write valid C/C++ programs.

eg →

```
#include <stdio.h>
int main()
{
    printf("Hello world");
    return 0;
}
```

→ Rules:-

1. statements end with semicolon(;))
2. Use curly braces { } for blocks.
3. variables must be declared before use.
4. keywords cannot be variable names.

* Conditions:-

→ conditions are used to make decisions in a program.

① if statement:-

→ Executes code only if the condition is true.

2. if-else statement:-

→ chooses between two blocks of code.

3. else-if Ladder:-

→ checks multiple conditions.

✓ Loops in C:-

→ Loops are used to repeat a block of code multiple times.

1. For loop :- Used when no. of iterations are known.

2. While Loop :- Checks the condition before executing.

3. Do-while Loop :- Executes at least once, then checks the condition.

42. Q.

WAP in C to print

```
*****
* * * * *
* * * * *
* * * * *
* * * * *
```

Code:-

Soln)

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```
    for(i = 1; i <= 5; i++)
```

```
    {
```

```
        for(j = 1; j <= 5; j++)
```

```
        {
```

```
            printf(" * ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

O/P →

```
*****
* * * * *
* * * * *
* * * * *
* * * * *
```

↓

43. Q.

WAP in C to print

```
1 → *
```

```
2 → * *
```

```
→ * * *
```

```
* * * *
```

```
* * * * *
```

42 Q.

WAP in C to print

```
*****
* * * * *
* * * * *
* * * * *
* * * * *
```

Solⁿ

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```
    for(i = 1; i <= 5; i++)
```

```
    {
```

```
        for(j = 1; j <= 5; j++)
```

```
        {
```

```
            printf("%c", '*');
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

O/P →

```
*****
* * * * *
* * * * *
* * * * *
* * * * *
```

43 Q.

WAP in C to print

```

      *
     * *
    * * *
   * * * *
  * * * * *

```


code:-

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```
    for (i = 1; i <= 5; i++)
```

```
    {
```

```
        for (j = 1; j <= i; j++)
```

```
        {
```

```
            printf(" * ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

O/P →

*

* *

* * *

* * * *

* * * * *

Q. 44. WAP in C to print

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

1 2 3 4 5

code:-

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int i, j;
```

```

for (i=1; i<=5; i++)
{
    for (j=1; j<=5; j++)
    {
        printf("%d", j);
    }
    printf("\n");
}
return 0;
}

```

OP →

```

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

```

45. Q. WAP in C to print

```

1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

Code:-

```

#include <stdio.h>
int main()
{
    int i, j;
    for (i=0; i<=5; i++)
    {
        for (j=0; j<=i; j++)
        {
            printf("%d", j);

```



```

    printf(" ");
}
    printf("\n");
} return
return 0;
}

```

O/P →

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Q. 46. WAP in c to print

N

```
#include <stdio.h>
```

```
int main()
```

```
{ int i, j;
```

```
for(i=1; i<=5; i++)
```

```
{ for(j=1; j<=5; j++)
```

```
{ if(i==1 || i==n || j==1 || j==n)
```

```
{ printf("*");
```

```
}
```

```
else printf(" ");
```

```
}
```

```
}
```

```
printf("\n");
```

```
}
```

```
return 0;
```

```
}
```

```

* * * * *
*           *
*           *
*           *
* * * * *

```

Q.47 WAP in C to print

```

      *
    *  *
  *   *   *
 *   *   *   *
*   *   *   *   *

```

```
#include <stdio.h>
```

```
int main()
```

```
{ int i, j;
```

```
  for(i=1; i<=5; i++)
```

```
  { for(j=1; j<=5; j++)
```

```
    { printf(" ");
```

```
    }
```

```
    for(j=1; j<=2*i-1; j++)
```

```
    { printf("*");
```

```
    }
```

```
    printf("\n");
```

```
  }
```

```
  return 0;
```

```
}
```


48. Q.

WAP in C to print

```

* * *
 * *
*
```

```
#include <stdio.h>
```

```
int main()
```

```
{ int n = 3;
```

```
for (i = n; i >= 1; i--)
```

```
{
```

```
for (int j = 1; j <= i; j++)
```

```
{
```

```
printf(" * ");
```

```
}
```

```
printf("\n");
```

```
return 0;
```

```
}
```

8.

```

*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

code:-

```

#include <stdio.h>
int main()

```

```

{
    int n=5;
    for (int i=1; i<=n; i++)
    {
        for (int j=1; j<=i; j++)
        {
            printf("*");
        }
        printf("\n");
    }

```

```

    for (int i=n-1; i>=1; i--)
    {
        for (int j=1; j<=i; j++)
        {
            printf("\n");
        }
    }
    return 0;
}

```


Q.

```

      *
    * * *
  * * * * *
* * * * * * *
* * * * * * *
  * * * * *
    * * *
      *
  
```

Solⁿ →

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n = 5;
```

```
    // Upper part
```

```
    for (int i = 1; i <= n; i++)
```

```
    {
```

```
        for (int j = 1; j <= n - i; j++)
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
        for (int j = 1; j <= 2 * i - 1; j++)
```

```
        {
```

```
            printf(" * ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    // Lower part
```

```
    for (int i = n - 1; i >= 1; i--)
```

```
    {
```

```
        for (int j = 1; j <= n - i; j++)
```

```
        {
```

```
            printf(" ");
```

```
        }
```

```
for (int j = 1; j <= 2 * i - 1; j++)  
{  
    printf("%* ", j);  
}  
printf("\n");  
}  
return 0;  
}
```